

Gouki Minegishi

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EDUCATION

Ph.D. student in Computer Science , University of Tokyo	Apr 2025–Mar 2028
<i>Advisor:</i> Yutaka Matsuo	
M.S., Computer Science , University of Tokyo	Apr 2023–Mar 2025
B.S., Engineering , University of Tokyo	Apr 2019–Mar 2023

SELECTED PUBLICATIONS

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- [1] **Gouki Minegishi**, Hiroki Furuta, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo.
“Topology of Reasoning: Understanding Large Reasoning Models through Reasoning Graph Properties.”
Neural Information Processing Systems (NeurIPS 2025).
 - [2] **Gouki Minegishi**, Hiroki Furuta, Shohei Taniguchi, Yusuke Iwasawa, and Yutaka Matsuo.
“Beyond Induction Heads: In-Context Meta Learning Induces Multi-Phase Circuit Emergence.”
International Conference on Machine Learning (ICML 2025).
 - [3] **Gouki Minegishi**, Hiroki Furuta, Yusuke Iwasawa, and Yutaka Matsuo.
“Rethinking Evaluation of Sparse Autoencoders through the Representation of Polysemous Words.”
International Conference on Learning Representations (ICLR 2025).
 - [4] Hirohane Takagi*, **Gouki Minegishi***, Shota Kizawa, Issey Sukeda, Hitomi Yanaka. (*) Equal Contribution
“Interpreting Multi-Attribute Confounding through Numerical Attributes in Large Language Models”
The 14th International Joint Conference on Natural Language Processing and The 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AACL 2025).
 - [5] **Gouki Minegishi**, Yusuke Iwasawa, and Yutaka Matsuo.
“Bridging Lottery Ticket and Grokking: Understanding Grokking from Inner Structure of Networks.”
Transactions on Machine Learning Research (TMLR).
 - [6] Hiroki Furuta, **Gouki Minegishi**, Yusuke Iwasawa, and Yutaka Matsuo.
“Towards Empirical Interpretation of Internal Circuits and Properties in Grokked Transformers on Modular Polynomials”
Transactions on Machine Learning Research (TMLR).
 - [7] Shohei Taniguchi, Keno Harada, **Gouki Minegishi**, et al.
“ADOPT: Modified Adam Can Converge with Any β_2 with the Optimal Rate.”
Neural Information Processing Systems (NeurIPS 2024).
 - [8] Hakaze Cho, Haolin Yang, **Gouki Minegishi**, and Naoya Inoue
Mechanism of Task-oriented Information Removal in In-context Learning
Preprint
 - [9] Kohsei Matsutani, Shota Takashiro, **Gouki Minegishi**, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo
RL Squeezes, SFT Expands: A Comparative Study of Reasoning LLMs
Preprint

RESEARCH/WORK EXPERIENCE

- Researcher, Third Intelligence, Inc June 2025–present
 - Research on developing and analyzing new forms of intelligence beyond conventional AI, aiming to bridge human-like reasoning, perception, and action.
- Research Intern, Preferred Networks, Inc Aug 2024-Oct 2024
 - Research on in-context learning in foundation models, with knowledge conflicts.
- Chief AI Engineer, Matsuo institute, Inc Nov 2021–May 2025
 - Developed automated driving software, including 3D object detection with deep learning and integration of the World Model into Automated Driving.
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ACADEMIC HONORS

- Selected for the **"BOOST NAIS" Scholarship** Program for fostering advanced AI talents leading the next-generation intelligent society
- **Oral presentation** accepted at the ICLR 2025 Workshop on Building Trust in Language Models and Applications, 2025
- **Young Researcher Encouragement Award**, NLP 2025
- **Outstanding Presentation Award**, JSAI Annual Conference 2024
- **Oral presentation** accepted at the ICLR 2024 Workshop on Bridging the Gap Between Practice and Theory in Deep Learning

ACADEMIC ACTIVITY

- Co-organizer for [Mechanistic Interpretability Organized Session](#) at JSAI2025.
- Reviewer for Neural Information Processing Systems (NeurIPS2025)
- Reviewer for International Conference on Machine Learning (ICML2025)
- Reviewer for International Conference on Learning Representations (ICLR2026)
- Reviewer for Annual AAAI Conference on Artificial Intelligence (AAAI2026)